Video-based Human Action Quality Assessment

1 Task 1: Understanding the FineDiving Paper

1.1 Action Recognition vs. Action Quality Assessment

Action recognition identifies which action is being performed, while action quality assessment evaluates how well it is performed by predicting a score. Unlike recognition, which may require only a few frames, AQA requires understanding the entire sequence and how the movement is executed. In diving, for example, subtle differences in body position and execution affect the score, even when athletes perform the same type of dive. Recognition models may overlook these details because they are trained to identify actions rather than judge their quality.

1.2 Why Does Temporal Information Matter in AQA?

Temporal information is important because the quality of an action depends not only on what happens, but also on how the movement evolves over time. In diving, a performance consists of several steps, including take-off, flight, and entry. Two divers may perform the same type of dive but differ in their body position during flight or the quality of their entry into the water. These differences may be subtle and cannot always be captured by looking at isolated frames. Considering the order and progression of the action can therefore help the model predict its quality more accurately.

1.3 Limitations of Predicting Scores from Global Video Features

Many AQA methods extract features from the entire video to predict a score directly. However, combining the whole performance into a single representation may hide important details. Execution quality depends on small mistakes and movements that occur at different moments. If these details are not well preserved, the model may miss differences that affect the final score. Another potential risk is that a model may rely on the type of dive rather than fully evaluating its execution. In Table 3, adding the dive number to the I3D features increases Spearman's correlation from 0.8576 to 0.8721. This suggests that dive identity helps predict the score, but does not necessarily mean the model understands how well the dive was performed.

1.4 Advantages of Decomposing Actions into Temporal Phases

Dividing an action into consecutive steps allows the model to analyse different parts of a performance instead of treating the video as a single unit, which is helpful because the quality of an action is not based on one moment in time or a specific frame, but rather on a sequence of movements throughout time. In FineDiving, the authors use temporal annotations to identify these steps and compare them with corresponding steps in an exemplar video. This makes it easier to identify differences between performances and understand which parts may affect the score. The results in Table 3 support this approach: Spearman's correlation increases from 0.8576 with global features and regression to 0.8793 when temporal segmentation is added, and reaches 0.9203 with the full TSA module. This suggests that dividing a performance into steps can improve prediction, while also making comparisons more interpretable.

1.5 Two Scientific Challenges Not Fully Addressed

First, the model's interpretability is not fully validated. Although the authors use attention heatmaps to show which steps contribute to predictions, these examples do not prove that the model correctly identifies the errors responsible for a score. Moreover, the segmentation accuracy is relatively low (AIoU@0.75 around 30–34%), also using ground-truth boundaries improves Spearman's correlation by only about 0.01. This raises the question of whether accurate step segmentation is essential to the

model's performance.
Second, the method relies on knowing the number of transitions between action steps, and the authors use a fixed number of steps for all dives. This may not reflect the different structures of more complex actions. The evaluation also trains and tests on samples from the same dataset, so its ability to generalise to new competitions or settings remains unclear. Testing on different events and with varying numbers of steps would help establish how robust the method really is.

2 Task 2: Practical Implementation

2.1 Dataset Exploration

I use AQA-7 [2], focusing on action class 1 (diving) and the official Split-4 lists. The dataset contains 300 training and 70 test clips. I verified that sample IDs are unique, the splits do not overlap, and all listed videos exist.

Scores range from 21.6 to 102.6, with a mean of 78.6 and a standard deviation of 14.3. Most scores lie between 70 and 90 (61%), while 9.8% are at or below 60, indicating an imbalanced target distribution.

Table 1: Video properties

Property	Value
Number of clips	370
Frames per clip	103
Frame rate	30 FPS
Resolution	320×240
Duration	≈ 3.4 s

Figure 1 shows the score distributions and individual scores. Training and test scores have similar distributions (Kolmogorov–Smirnov $D = 0.058$, $p = 0.984$). However, the concentration of scores around 70–90 may favour predictions near the mean, motivating comparison with a mean-prediction baseline and reporting both MAE and Spearman correlation.

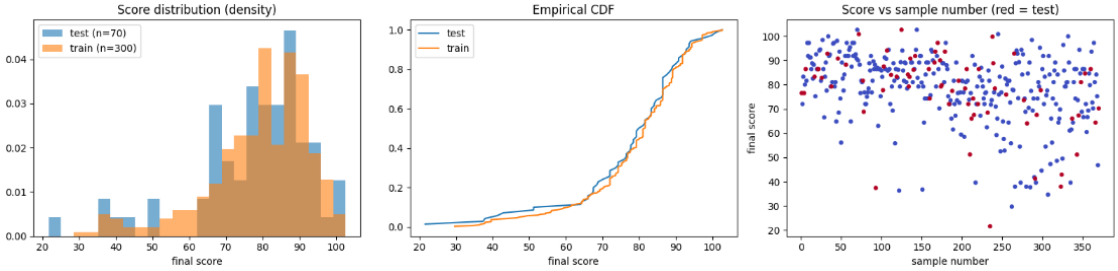


Figure 1: Score distributions for the training and test sets (histograms and ECDFs), and scores by sample number.

Other challenges are the small size of the dataset (only 240 clips are left for training after validation), the fact that only the final score is given (so the difficulty of a dive cannot be separated from how well it was executed), and many tied scores (only 121 distinct values for 370 clips). All clips have the same length, but the phases of the dive do not happen at the same frames in every clip.

2.2 Data Split and Evaluation

I keep the official 70 test clips as the test set. From the 300 official training clips, I hold out 60 for validation: I sort them by score and take every fifth clip, starting from a random position fixed by the seed. This makes sure the validation set also contains low-scoring dives (a first attempt, sampling within score groups, had no validation dive below 52.8). The remaining 240 clips are used for training. All choices (hyperparameters, early stopping) are made on the validation set, and the test set is used only once, at the end.
Both metrics have limitations for AQA. MAE is easy to read, in score points, but with imbalanced scores a model can get a fairly low MAE by predicting values close to the mean. Spearman correlation is not fooled by this, but it ignores the actual values: a model can rank dives correctly while being several points off. It is also affected by ties and is noisy on 70 clips. I therefore report both, compare against the mean predictor, and give 95% bootstrap confidence intervals in the notebook.

2.3 Video Features

I do not train a video network. Each clip is passed once through a frozen I3D network [3] (PyTorch port by piergiaj). Following FineDiving [1], frames 0–95 are cut into 9 snippets of 16 frames, one every 10 frames, and I3D turns each snippet into 1,024 numbers. Each dive is therefore described by a 9×1024 array. **All the models below use exactly the same features, so differences between them only come from the models.**

2.4 Baseline Model

I first implemented two simple baselines to check whether the video features contain useful information for predicting dive quality.
The first baseline (R0) always predicts the mean score of the training set, regardless of the input video. It provides a simple reference point. The second baseline (R1) uses Ridge regression, a linear model that predicts the score from the video features. For R1, I average the features of the nine video snippets into a single vector, then use this vector to predict the score. The regularization parameter is selected using the validation set.

Table 2: Baseline results on the test set.

Model	**Spearman ρ**	**MAE**
R0: Mean predictor	–	11.135
R1: Ridge regression	0.755	9.277

Ridge regression achieves a lower MAE than the mean predictor, suggesting that the video features provide useful information for predicting scores. Spearman correlation is undefined for R0 because it gives the same prediction to every video so they can't be ranked.

2.5 Temporal Model

The baseline averages all nine snippets, which removes information about their order. However, the quality of a dive may depend on how it progresses through take-off, flight and entry. I therefore compared five ways of combining the snippet features while keeping the same pretrained I3D features as input.
Figure 2 shows the general architecture. A projection layer reduces each snippet's 1,024 features to a smaller representation of 64 or 128 values. A temporal aggregator combines the nine snippets, and a final prediction head outputs one score.

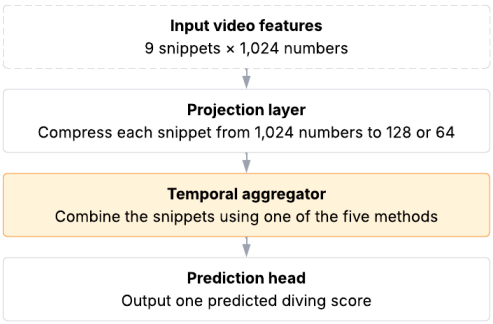


Figure 2: Architecture of the models used for dive quality prediction.

The five aggregation methods are:

- **Mean pooling:** averages all nine snippets, without explicitly using their order.
- **Three segments (seg3):** divides the snippets into three consecutive groups and combines their average features, preserving a coarse representation of the video progression.
- **Temporal convolutional network (TCN):** learns patterns between nearby snippets.
- **GRU:** processes the snippets as a sequence to learn relationships over time.
- **Attention (attn):** assigns different weights to the snippets before combining them, allowing some parts of the video to contribute more to the prediction than others.

I selected the hyperparameters and the best training epoch using validation Spearman correlation. Each model was then trained with five different random seeds. For the final test evaluation, I averaged the five runs' predictions for each video.

Table 3: Validation and test results for the temporal models.

Model	Val. ρ	Test ρ	Test MAE
Mean pooling	0.580	0.776	7.308
Three segments	0.585	0.731	6.506
TCN	0.593	0.749	9.433
GRU	0.598	0.735	6.859
Attention	0.615	0.766	6.236

I used validation Spearman correlation to select models for further analysis. Mean pooling served as a simple baseline, combining all nine snippets without explicitly modelling their order. I also selected attention because it achieved the highest validation correlation (0.615). I then evaluated all five temporal models on the test set, which was used only for final evaluation.
Attention achieves the lowest test MAE (6.236), while mean pooling achieves the highest test Spearman correlation (0.776). The three-segment model also achieves a low MAE (6.506). These results show that the best model depends on the metric: attention predicts numerical scores more accurately on average, while mean pooling ranks dives better. However, no temporal model clearly improves ranking over mean pooling, and the results should be interpreted cautiously because the test set contains only 70 videos.

2.6 Uncertainty in Test Results

To account for the small test set, I computed 95% bootstrap confidence intervals. Bootstrap resampling repeatedly selects 70 videos from the test set, allowing the same video to be selected more than once. By recalculating the metrics across 2,000 resamples, I estimated how much the results could vary depending on which videos were evaluated. The 95% confidence interval is the range between the 2.5th and 97.5th percentiles of these estimates. The results for the two best models by MAE and the mean-pooling baseline are summarised below.

Table 4: Test performance with 95% bootstrap confidence intervals.

Model	Spearman ρ (95% CI)	MAE (95% CI)
Mean pooling	0.776 [0.638, 0.868]	7.308 [5.96, 8.83]
Three segments	0.731 [0.583, 0.834]	6.506 [5.25, 7.91]
Attention	0.766 [0.621, 0.861]	6.236 [5.16, 7.40]

Although attention and three-segment pooling achieve lower MAE than mean pooling, their confidence intervals overlap. This does not establish that the models perform equally, but the observed differences alone are insufficient to show a clear improvement. In Task 3, I therefore compare their predictions with mean pooling using a paired bootstrap test, which measures the difference in MAE on the same resampled videos.

3 Task 3: Experimental Study (Option A)

Question. Does explicit temporal modelling improve quality prediction compared with a global video representation?
In Task 2, attention and the three-segment model had a lower MAE than mean pooling, but the 95% confidence intervals of the models overlapped. Comparing separate intervals is a weak test: part of their width comes from which dives happen to be in the test set, and a hard set of dives is hard for every model.
Method: paired bootstrap I draw 70 dives at random, with replacement, from the 70 test dives, and compute $\Delta\text{MAE} = \text{MAE(model)} - \text{MAE(mean pooling)}$ and $\Delta\rho$ in the same way. Since both models are scored on exactly the same dives, the difficulty of the sample cancels out. I repeat this 2,000 times and take the middle 95% of the differences as the confidence interval. I use the same resamples for every comparison and the 5-seed averaged predictions from Task 2, so nothing is retrained.
Since attention achieved the best validation ρ, the main hypothesis was that it would reduce MAE compared with mean pooling. Three-segment pooling was included as a secondary temporal comparison. I found that attention reduces test MAE from 7.31 to 6.24 points ($\Delta\text{MAE} = -1.07$, 95% paired-bootstrap

CI [−2.00, −0.23]), with lower error in 99.4% of bootstrap resamples. Three-segment pooling also reduces MAE to 6.51 (ΔMAE = −0.80, 95% CI [−1.56, −0.03]), with lower error in 97.8% of resamples. However, neither model shows a clear improvement in Spearman correlation: the differences are −0.010 (95% CI [−0.060, 0.033]) for attention and −0.045 (95% CI [−0.113, 0.005]) for three-segment pooling.
Interpretation (hypothesis) Attention may predict scores more accurately without better distinguishing stronger from weaker dives. All models over-predict rare low scores; attention has the smallest error on these cases.
Limitations

- **Confounding factors:** Attention also uses learned weights, positional embeddings and different hyperparameters. shuffling the order of the nine video snippets would be needed to isolate the effect of temporal order.
- **Small test set:** Only 70 dives are tested, and the results may differ on another set of videos. The same athletes may also appear in both training and test sets. Bootstrap estimates uncertainty due to the test videos, but not differences caused by training randomness.
- **Noisy selection:** Validation contains only 60 dives, and its Spearman correlation is lower than test correlation, making model selection uncertain.
- **Temporal limitations:** Using only nine snippets may miss small movements.

4 Task 4: Research Proposal

Recognising difficulty or assessing execution?

Problem. In diving, the final score is the degree of difficulty (DD) multiplied by the sum of the judges' execution marks. DD depends only on which dive is attempted. So part of the final score can be predicted just by recognising the dive, without judging how well it was performed. Most AQA methods, including FineDiving's TSA, predict the final score directly, and Spearman correlation or MAE on the final score cannot tell these two abilities apart.
Motivation. In my experiments on AQA-7, only final scores were available, so I could not tell whether $\rho \approx 0.75$ came from judging execution or from recognising hard dives. FineDiving [1] provides DD and the judges' scores, which makes this question testable. An important question is whether current AQA models genuinely assess execution quality or partly rely on dive difficulty, and how well they distinguish performances of similar difficulty. The same problem exists in other judged sports, such as gymnastics and figure skating.
Hypotheses: H1: a large part of the reported ρ can be obtained from difficulty alone, and ρ measured within a single dive type is much lower than the overall ρ.
H2: a model that predicts execution separately and multiplies it by DD reaches a similar accuracy on the final score, but predicts execution better and ranks dives of the same type better.
Proposed method. The same frozen I3D features as in my experiments, with two heads: (i) a *difficulty head* that classifies the dive type and converts it to DD with the official table (DD labels are also given in FineDiving), and (ii) an *execution head* that predicts the sum of the judges' marks. The final score is their product. As an optional extension, the execution head could use crops centred on the diver (person detector and tracker), since execution is mostly about body position.
Experiments

1. **Difficulty-only baseline:** predict DD × the mean training execution score, with true and with predicted DD, and compare its ρ with direct regression. H1 is refuted if this baseline has a low ρ and direct models keep a similar ρ within dive types.
2. **Direct vs. factorised:** report ρ and MAE on the final score, on the execution score, and the average ρ within dive types. H2 is refuted if the paired intervals on the last two contain zero.
3. **Ablations:** true vs. predicted DD; full frame vs. diver-centred crops.
4. **Generalisation:** repeat on other datasets, which also provide DD and the judges' scores.

Expected contribution and risks. The main contribution is an evaluation protocol that separates recognising the action from assessing it, and that applies to any AQA method. Risks: Incorrect dive-type predictions can lead to incorrect final scores. Execution scores may vary only slightly between dives of the same type, making correlation harder to measure reliably.

References

[1] J. Xu, Y. Rao, X. Yu, G. Chen, J. Zhou, and J. Lu, "FineDiving: A Fine-Grained Dataset for Procedure-Aware Action Quality Assessment," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 2949–2958. doi: 10.1109/CVPR52688.2022.00296. Available: https://openaccess.thecvf.com/content/CVPR2022/html/Xu_FineDiving_A_Fine-Grained_Dataset_for_Procedure-Aware_Action_Quality_Assessment_CVPR_2022_paper.html

[2] P. Parmar and B. T. Morris, "Action Quality Assessment Across Multiple Actions," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2019, pp. 1468–1476. doi: 10.1109/WACV.2019.00161. Available: https://doi.org/10.1109/WACV.2019.00161

[3] J. Carreira and A. Zisserman, "Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 6299–6308. doi: 10.1109/CVPR.2017.502. Available: https://openaccess.thecvf.com/content_cvpr_2017/html/Carreira_Quo_Vadis_Action_CVPR_2017_paper.html